

CS201 Midterm 1 Part B

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I. QUESTION (LINKEDLIST) (15 POINTS)

Write a method which returns true if the single linked list only contains triplicates, that is, every datum (number) occurs only three times. Important warning, the triplicate elements may not be adjacent.

```
boolean containsOnlyTriplicates()
```

Your method should have a time complexity of $\mathcal{O}(N^2)$. You are not allowed to use any single linked list methods, just attributes, constructors, getters and setters.

II. QUESTION (DOUBLYLINKEDLIST) (15 POINTS)

Write the method

```
void reverse()
```

for reversing a doubly linked list.

Your method should have a time complexity of $\mathcal{O}(N)$. You are not allowed to use any extra data structures. You are not allowed to use any linked list methods, just attributes, constructors, getters and setters.

III. QUESTION (STACK) (15 POINTS)

Write a function that inserts a new integer after the largest element of the stack. Write the function for array implementation. You are not allowed to use any stack methods, just attributes, constructors, getters and setters.

```
void insertAfterLargest(int s)
```

Contents of the stack before inserting 6:

1 8 7 3

Contents of the stack after inserting 6:

1 8 6 7 3

IV. QUESTION (STACK) (15 POINTS)

Write the method

```
void removeOddIndexed()
```

which removes only the odd indexed (1, 3, ...) elements from the stack. The first element has index 1. You are only allowed to use pop, push, isEmpty functions. You should use a single external stack. You should use linked list implementation of stack.

Contents of the stack before:

1 2 0 4 6 9

Contents of the stack after:

2 4 9